
THEO OLSEN

Vassar College
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EDUCATION

- 2023 - 2027 B.A. Physics & Astronomy, Vassar College, Poughkeepsie, NY
Correlate: Applied Mathematics
- 2019 - 2023 Regents Honors Diploma, LaGuardia High School, New York, NY

RESEARCH SPECIALIZATIONS AND INTERESTS

I am a physics & astronomy student interested in research on brown dwarfs, circumstellar disks, exoplanets, and planet formation.

PUBLICATIONS

- 2026 Suarez, Cruz, Faherty, Kiman, Metchev, Oliveros-Gomez, Alvarado III, **Olsen**, *et al.* “*SEDA: Spectral Energy Distribution Analyzer for Forward Modeling and Empirical Analysis of Ultracool Objects*”
Journal of Open Source Software (in-review)
<https://github.com/openjournals/joss-papers/blob/joss.10158/joss.10158/10.21105.joss.10158.pdf>
- 2024 **Olsen** “*Disk-tective Work: The Link Between Inner Disk Gas and Terrestrial Planet Formation*”
Keck Northeast Astronomy Consortium: 35th Annual Undergraduate Symposium Proceedings.
<https://drive.google.com/file/d/15XKWieIpFxSj1ILWEHOvdJeyJL5IDRhFg/view>

CONFERENCES

- 2025
April **Olsen, T.** “*Disk-tective Work: The Link Between Inner Disk Gas and Terrestrial Planet Formation*”
Poster presentation at the Vassar-West Point Joint Symposium, Vassar College, NY.
- 2024
September **Olsen, T.** “*Disk-tective Work: The Link Between Inner Disk Gas and Terrestrial Planet Formation*”
15-minute oral presentation at the 35th Annual Keck Northeast Astronomy Consortium Symposium, Colgate University, NY.

- 2024
September **Olsen, T.** “*Disk-tective Work: The Link Between Inner Disk Gas and Terrestrial Planet Formation*”
Poster presentation at the Undergraduate Research Summer Institute Symposium, Vassar College, NY.
- 2023
July Li, W., **Olsen, T.**, Yan, J. “*Giants vs. Dwarfs: Most Abundant Star in the Universe*”
Poster presentation at the 10th Annual NYC Science Research Mentorship Colloquium, American Museum of Natural History, NY
- 2024
May Participant in the “New York Area Exoplanets Meeting”,
Hosted by the Center for Computational Astrophysics, Flatiron Institute, NY

RESEARCH EXPERIENCE

- 2026
May - Present **Astronomy Research Intern**
American Museum of Natural History, Department of Astrophysics

Added support for evolutionary modeling and inclination estimation to SEDRA python package for analyzing ultracool dwarfs. Conducting research exploring the relationship between brown dwarf rotation and silicate cloud structure using JWST data.
Research Mentor: Dr. Genaro Suárez
- 2026
April - Present **Observatory Assistant**
Vassar College, The Class of 1951 Observatory

I help host weekly open nights at the observatory, where we teach community members basic observational astronomy. I am responsible for operating the 32” reflecting telescope at the Class of 1951 Observatory, as well as reducing and analyzing observational data. We participate in the TESS Follow-Up Observing Program (TFOP), contributing ground-based photometric observations of planet candidates from NASA's TESS mission.
Supervisors: Dr. Colette Salyk, Steve Dittmar
- 2025
May - September **Astronomy Student Researcher**
Vassar College, Department of Astronomy

Worked with a professor to write and test new code for identifying density clumps in a magnetohydrodynamic simulated CGM. Used bash with a high power computing cluster to run code in parallel.
Research Mentor: Dr. Ed Buie II

Astronomy Student Researcher
 2024 Vassar College, Undergraduate Research Summer Institute (URSI)
 May -
 September Worked under a postdoctoral researcher to classify protoplanetary disk substructure based on ALMA observations and explore connections between their structure, characteristics from literature, and spectroscopy. Published a conference proceedings paper on the topic, and presented my work at three symposiums.
 Research Mentor: Dr. Annie Dickson-Vandervelde

Other:
 2025 **Undergraduate Research Assistant**
 January - May Vassar College, Department of Philosophy
 Assisting with developing a new Intermediate Logic course, grading student problem sets, and holding tutoring hours to provide support to Logic students.
 Instructor: Dr. Kathryn Pendoley

AWARDS

2026 Vassar Internship Grant Award, Supporting Research at AMNH

2025 International University Physics Competition, Bronze Medalist

2025 Vassar College Concerto Competition Winner

2023 LaGuardia Alumni & Friends Student Musician Award

2021 HOK Architecture, 2nd Place in CIRT National Design Competition

RELEVANT COURSEWORK

ASTR 330 Extrasolar Planets

ASTR 322 Galaxies & Galactic Structure

ASTR 320 Astrophysics of The ISM

ASTR 240 Observational Astronomy

ASTR 230 Planetary/Space Physics

ASTR 220 Stellar Astronomy

ESCI 383 Planetary Geology

PHYS 320 Quantum Mechanics I

PHYS 261 Solid State Physics & Ultrafast Optics

PHYS 240 Electromagnetism I

PHYS 245 Statistical Mechanics & Thermodynamics

PHYS 210 Classical Mechanics

PHYS 203	Experimental Physics
PHYS 200	Modern Physics
MATH 315	Topics in Math Modeling
MATH 228	Ordinary Differential Equations
MATH 221	Linear Algebra
MATH 220	Multivariable Calculus
MATH 127	Calculus IIB: Sequences & Series
CMPU 101	Computer Science I

SERVICE

<i>2024 - Present</i>	Co-President , Physics & Astronomy Journal Club, Vassar College
<i>2025 - Present</i>	Research Representative , Physics & Astronomy Majors Committee
<i>2024 - Present</i>	President, Founder , Mahagonny Ensemble, Vassar College
<i>2023 - Present</i>	Executive Board Member , Camerata Ensemble, Vassar College
<i>2023 - Present</i>	Co-principal Flute , Vassar College Orchestra
<i>2024 - Present</i>	Professional Cheesemonger , Cheese Plate Brooklyn
<i>2024 - Present</i>	Member , Vassar-Chadwick Chess Club
<i>2024 - 2026</i>	Usher , Vassar College Music Department

COMPUTER SKILLS

High:	Git, Python, bash, LaTeX, Overleaf, Jupyter Notebook, Matplotlib, SciPy, NumPy, Microsoft Excel, Google Colab, Powerpoint
Intermediate:	SIMBAD, Vizier, TOPCAT, R

REFERENCES

References available upon request.